Text-message reminders and hand hygiene among nurses: a randomised controlled trial

Priya Sharma1*, Rahul Verma2

1 Department of Community Medicine, Lakeview Medical College, Pune, India. 2 School of Nursing, Lakeview University, Pune, India.

*Corresponding author: priya.sharma@lakeview.edu.in; ORCID 0000-0002-1825-0097

Abstract

Background: Reminders may improve hand hygiene. Methods: In this randomised controlled trial, 160 nurses received text reminders or usual care for 12 weeks. Results: Compliance rose by 15%. Conclusion: Reminders help.

Keywords

hand hygiene; reminders; nurses; randomised controlled trial

Introduction

Hand hygiene compliance is low in many hospitals [1]. Reminders can change behaviour [2].

Methods

This parallel-group randomised controlled trial was conducted between January 2024 and June 2024 at Lakeview Medical College and is reported according to CONSORT. It was registered with the Clinical Trials Registry of India (CTRI/2024/06/071234).

A total of 160 nurses (n = 160) took part. Inclusion criteria were permanent ward staff; exclusion criteria were planned leave. The sample size was calculated to give 80% power.

The randomisation sequence was computer-generated in blocks of four using custom scripts written in Python. Allocation was concealed in sealed opaque envelopes, and outcome assessors were blinded. The primary outcome was hand hygiene compliance at 12 weeks.

Statistical analysis used chi-square tests in SPSS version 26.

The study was approved by the Institutional Ethics Committee (approval no. IEC/2023/198, dated 4 December 2023) and followed the Declaration of Helsinki. Written informed consent was obtained from all participants.

Results

Of the 160 nurses, 128 (80.0%) were women. The CONSORT flow diagram is shown in Figure 1. Compliance rose more with reminders (Table 1) [3].

Discussion

Similar effects were seen in earlier trials [4].

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Conflicts of interest

The authors declare no conflicts of interest.

Author contributions

PS: Conceptualization, Methodology, Writing – original draft. RV: Formal analysis, Investigation, Writing – review & editing.

Data availability statement

The anonymised data are deposited in Zenodo (https://doi.org/10.5281/zenodo.7654321).

Declaration of generative AI use

ChatGPT (OpenAI) was used to improve the language of the manuscript; the authors checked all text.

Figure 1. CONSORT flow diagram, adapted from Author3 et al. [3].

Table 1. Compliance by group.

Variable

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